

MotionApps for IAM-20680/20680HP/20680HT Release Notes

Version 13.1.1 - MP

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1. OVERVIEW

1.5. RELEASE PURPOSE

This release goal is to provide an IAM-20680 Linux/Android stack with improved code quality.

1.6. AUDIENCE

This release is intended for TDK-InvenSense Customers & TDK-InvenSense field engineers who want to use TDK-InvenSense HW/SW sensor system.

2. RELEASE CONTENT

2.5. SOFTWARE MODULE INCLUDED

- Linux driver version 10.0.7
@ *kernel/invensense/linux-icm20xxx*
- Android MA Sensors HAL version 13.1.2
@ *aosp/hardware/invensense/sensors*
- Android Sensors Multi-HAL 2.1 wrapper version 1.0.0
@ *aosp/hardware/invensense/subhalv2_1*
- Test applications for Android
@ *aosp/hardware/invensense/tools*
- Optional: Android patch for Heading sensor support
@ *patches/hardware/interfaces*
- Documentation
@ *README*

2.6. SUPPORTED FEATURES

2.6.1. Android Sensor Types

- Gyroscope
- Gyroscope Uncalibrated
- Accelerometer
- Accelerometer Uncalibrated
- Magnetometer
- Magnetometer Uncalibrated
- Pressure
- Orientation
- Rotation Vector
- Game Rotation Vector
- Geomagnetic Rotation Vector
- Linear Acceleration
- Gravity
- Heading

2.6.2. HW based features

- Gyroscope raw data
- Accelerometer raw data

2.6.3. SW Features

These features are supported by a library inside Sensors HAL:

- Gyroscope calibration
- Accelerometer calibration
- Magnetometer calibration
- 6-axes (A + G) sensor fusion
- 6-axes (A + M) sensor fusion
- 9-axes (A + G + M) sensor fusion

The Sensors HAL supports Direct Report Mode using shared-memory.

2.7. TEST PLATFORM

- Qualcomm RB5 board
 - AOSP 13.0.0_r1 64 bits
 - Linux kernel 5.15.41
 - Sensors HAL AIDL
 - IAM-20680 on I2C
- CTS 13.0_r1: [PASS](#)
- VTS 13.0_r1: [PASS](#)

2.8. PORTING

Please refer to README file enclosed in the release package and a separate Porting Guide document.

2.9. CHANGELOG

There are following updates related to IAM-20680 until release MA 13.1.0.

ID	Module	Title	Note
MOT-173	Driver	Fix code quality issue	Fix issues reported by checkpatch and sparse
MOT-172	HAL	Fix coverity issues	Fix issues reported by coverity tool

2.10. KNOWN LIMITATIONS

Some CTS test cases fail when HiFi Sensor and VR feature are enabled on the platform because of hardware limitations.

1. FIFO size
HiFi Sensor requires large HW FIFO such as 3000 events for Accelerometer.
2. Significant Motion sensor type
This sensor type is not supported by the hardware.

2.11. ADDITIONAL INFORMATION

Engineering tests have been performed for this release on following platforms.

- AOSP 13.0.0_r1 + Linux 5.15 on Qualcomm RB5 with IAM-20680 on I2C
 - CTS 13.0_r2: [PASS](#)
 - VTS 13.0_r1: [PASS](#)

3. **REVISION HISTORY**

REVISION DATE	REVISION	DESCRIPTION
05/26/2023	1.0	Initial release

4. **COMPLIANCE DECLARATION DISCLAIMER**

InvenSense believes the environmental and other compliance information given in this document to be correct but cannot guarantee accuracy or completeness. Conformity documents substantiating the specifications and component characteristics are on file. InvenSense subcontracts manufacturing and the information contained herein is based on data received from vendors and suppliers, which has not been validated by InvenSense.